Chinese CEFR Labels Inherit an Ambiguity: 46% of Texts Change Band with the Source Standard

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Abstract

Multilingual CEFR resources now span thirteen languages (Imperial et al., 2025); Chinese is not among them. The obvious route in is HSK, the Chinese proficiency scale, which is routinely mapped onto the CEFR's six bands. That route is less well defined than it looks. "HSK 3.0" names two official documents — the GF0025-2021 national grading standard and the examination syllabus in force since July 2026 — which assign different levels to 41.5% of the vocabulary they share (Serrano, 2026). We show the ambiguity propagates: grading 102 graded readers against one document rather than the other moves **46.1%** of them into a different CEFR band under the usual one-to-one correspondence, and **44.1%** under a compressed correspondence that assumes HSK 6 ≈ B2. It replicates on a disjoint 30-text held-out split (46.7%, 43.3%). The movement is not noise: **42 of 47** texts move *upward*, and on the held-out split all 14 do. Choosing the wrong document does not add error, it shifts the distribution. We argue that any Chinese entry in a CEFR corpus must record which Chinese standard its labels derive from, and that no correspondence table can repair the problem, because the ambiguity sits upstream of the mapping.

1 The gap, and why it is awkward to fill

Imperial et al. (2025) assemble 494,491 CEFR-labelled texts across thirteen languages and standardise them into one schema. Chinese is absent. This is not an oversight of scale — Chinese is among the most-studied second languages in the world — but a consequence of the inclusion criteria, which require a permissive licence, CEFR labels produced or validated by domain experts, and human authorship.

The natural way to produce Chinese CEFR labels at scale is indirect: grade text against HSK, then map HSK levels onto CEFR bands. Learner-facing material does this constantly. The mapping is not official — China's own standard declines to align itself to the CEFR — but it is ubiquitous, and it is the implicit basis on which Chinese material is described as "B1" or "C1" in practice.

This note reports an obstacle that sits before the mapping, and that a better mapping cannot fix.

2 Two documents named HSK 3.0

The name "HSK 3.0" is applied to two distinct official publications:

- **《国际中文教育中文水平等级标准》** (GF0025-2021), the national grading standard, in force since 1 July 2021. Nine levels, 11,092 words, 3,000 characters.
- **新版 HSK 考试大纲**, the examination syllabus, published November 2025 and in force since July 2026. 10,896 words, 3,088 recognition characters, and a separate 1,200-character writing inventory.

Serrano (2026) compares them directly: of the 9,674 words the two share, 41.5% sit at different levels; of the 2,945 shared characters, 40.7% do. Grading a corpus of authentic graded readers against one rather than the other changes the assigned HSK level of 48.0% of texts. Tools and reference sites that report "HSK 3.0" levels overwhelmingly do not say which document they used.

The question this note asks is narrow: does that disagreement survive projection onto the CEFR's coarser six-band scale, or does coarsening wash it out?

	Readers (n=102)	Held-out (n=30)
HSK level changes	49 (48.0%)	14 (46.7%)
CEFR band, naive	47 (46.1%)	14 (46.7%)
upward	42	14
downward	5	0
CEFR band, compressed	45 (44.1%)	13 (43.3%)
upward	42	13
downward	3	0

Table 1: Texts whose band differs depending on which HSK document was used. "Upward" means the 2025 syllabus placed the text higher than the 2021 standard did.

3 Method

Corpus. 102 graded readers spanning six difficulty shelves, with per-word segmentation authored rather than produced by a segmenter, plus a disjoint 30-text held-out split. Both are released CC BY 4.0 with a datasheet. We grade the authored segmentation directly; re-tokenising the raw strings with a segmenter yields materially different numbers and is not comparable to the published figures.

Grading. Each text is graded twice with the same open implementation (Serrano, 2026), once against GF0025-2021 and once against the 2025 syllabus. A text's level is the lowest level whose cumulative character inventory covers 95% of its running characters, the minimal coverage threshold from Laufer and Ravenhorst-Kalovski (2010). Nothing about the procedure differs between the two runs except the inventory.

Correspondences. We use two, and report every figure under both. The **naive** correspondence is the one-to-one reading ubiquitous in learner material: HSK n maps to the nth CEFR band, with HSK 3.0's combined levels 7–9 mapped to C2. The **compressed** correspondence encodes the objection practitioners raise — that vocabulary coverage overstates communicative range, so HSK 6 lands nearer B2 than C2 — by mapping $\{1,2\}\to$A1, $3\to$A2, $4\to$B1, $\{5,6\}\to$B2, $\{7,8\}\to$C1, $9\to$C2.

Neither is authoritative. That is the point of using both: a finding that holds under both does not rest on the choice.

	A1	A2	B1	B2	C1	C2
2021 basis	3	28	32	18	14	7
2025 basis	5	10	29	32	19	7

Table 2: Band distribution of the same 102 texts under the naive correspondence. The A2 population falls by 64% and the B2 population rises by 78%; the corpus is not relabelled at random, it is relabelled harder.

4 Results

Table 1 gives the result. Coarsening to six bands does not wash the disagreement out: 46.1% of texts change band under the naive correspondence, against 48.0% that change HSK level. Compressing the correspondence — which merges adjacent HSK levels and so should absorb single-level shifts — reduces it only to 44.1%. The pattern replicates on the held-out split at 46.7% and 43.3%.

The direction is the more consequential finding. Of the 47 texts that move under the naive correspondence, 42 move upward; on the held-out split, all 14 do. Table 2 shows the effect on the corpus as a whole: the A2 population falls from 28 to 10 while B2 rises from 18 to 32. A researcher who picks the wrong document does not acquire noisy labels. They acquire a corpus that is systematically mis-levelled in one direction, which is the failure mode least likely to be caught by inspection and most likely to bias a model trained on it.

5 What this implies for multilingual CEFR resources

The provenance field is not optional. The Imperial et al. (2025) schema records source dataset, language, granularity, production category and licence. For a Chinese entry this is insufficient: two entries could share every one of those values, be produced by the same procedure over the same texts, and still disagree on the band of nearly half of them. Chinese needs the source standard, with its version, recorded as data. We would expect the same to apply wherever a language's CEFR labels are derived through a national framework rather than assigned directly.

No correspondence table fixes it. It is tempting to read this as an argument for a better HSK–CEFR mapping. It is not. Both correspondences we test are applied identically to both gradings; the disagreement is generated before the mapping is consulted and passes through it. A third correspondence would behave the same way.

Derived labels are not expert labels. We are explicit that the labels here are algorithmic projections of a coverage threshold, not expert-assigned CEFR levels, and so do not meet the gold-standard criterion of Imperial et al. (2025). The contribution is not a proposed dataset entry. It is a constraint that a future entry will have to satisfy.

6 Limitations

The corpus is LLM-assisted, disclosed in its datasheet, and pedagogically constrained; it is not a sample of naturally occurring Chinese and would not meet the human-authorship criterion of Imperial et al. (2025). Its levels are ours, not experts'. Coverage is necessary but not sufficient for comprehension: grammar, register and world knowledge are unmodelled, and the 95% threshold is imported from word-level English research rather than established for L2 Chinese characters. Neither correspondence we use is official, and we make no claim that either is correct — only that the finding does not depend on which is chosen. Finally, the 2025 syllabus takes effect in July 2026, so its practical consequences are not yet observable in learner outcomes.

7 Availability

The grading implementation is MIT licensed and installable as `pip install hsk30`; the corpus and held-out split are CC BY 4.0. Every figure in this note is regenerated by `scripts/cefr_mapping.py` in the same repository, which prints the table above from the corpus and the two inventories.

Ethics and data statement

The corpus contains no personal data and describes invented characters in everyday situations; its LLM-assisted authorship is disclosed in its datasheet and again in the limitations above. Word and character lists derive from published national standards. No human subjects were involved. An intended application is a Chinese-learning website operated by the author.

References

Joseph Marvin Imperial, Abdullah Barayan, Regina Stodden, Rodrigo Wilkens, Ricardo Muñoz Sánchez, Lingyun Gao, Melissa Torgbi, Dawn Knight, Gail Forey, Reka R. Jablonkai, Ekaterina Kochmar, Robert Joshua Reynolds, Eugénio Ribeiro, Horacio Saggion, Elena Volodina, Sowmya Vajjala, Thomas François, Fernando Alva-Manchego, and Harish Tayyar Madabushi. 2025. UniversalCEFR: Enabling open multilingual research on language proficiency assessment. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 9703–9755, Suzhou, China. Association for Computational Linguistics.

Batia Laufer and Geke C. Ravenhorst-Kalovski. 2010. Lexical threshold revisited: Lexical text coverage, learners' vocabulary size and reading comprehension. *Reading in a Foreign Language*, 22(1):15–30.

Alvaro Serrano. 2026. Which HSK 3.0? two official documents, and half of all grading decisions change. Preprint.